V4.2 Toy Model 2A
Two-State Reciprocal Projection Test

Rev A - Exploratory Mathematical Compatibility Report

Date: 2026.05.25 | *Prepared for continuation of the 6D Projection Framework*

Status: This chapter begins the quantum-side formalization. It tests compatibility, not proof. The target is a minimal two-state projection model, not a full derivation of Quantum Mechanics.

1. Purpose

Toy Model 1D closed the first graph-based operational-geometry chapter. Toy Model 2A starts the quantum-side chapter by testing the smallest possible projected quantum structure: a two-state system.

The working question is narrow: can a hidden three-dimensional temporal-relational orientation be mapped into a two-state observable quantum-like amplitude structure while preserving probability normalization, Born-rule compatibility, unitary rotation compatibility, and no-signalling?

This report does not claim to derive the Born rule. It defines a candidate reciprocal projection operator and checks whether it survives first-contact constraints.

2. Starting architecture

• Parent architecture: a 6D structure, interpreted as three spatial and three temporal degrees of freedom.

• Observable sector: 3S + 1T, the classical observer-accessible sector.

• Hidden relational sector: 3T + 1S, the proposed temporal-relational sector.

• Reciprocal projection target: hidden 3T relational structure maps into observable 1T outcome restriction.

• First test case: a two-state quantum system, also called a qubit.

3. Core notation

We introduce the projection symbol Pi:

```
psi_obs = Pi(Psi_3T)
```

• Psi_3T = hidden temporal-relational state.

• Pi = projection operator.

• psi_obs = observer-accessible two-state quantum-like state.

For the two-state test, the observable state is written:

```
|psi> = alpha |0> + beta |1>
```

• |0> and |1> are basis states. A basis state is one of the basic possible outcomes.

• alpha is the amplitude for outcome 0.

• beta is the amplitude for outcome 1.

• An amplitude is not directly a probability. The probability is calculated from the squared magnitude of the amplitude.

4. Candidate hidden 3T object

For the minimal test, the hidden relational state is represented by a normalized direction vector n in a three-dimensional hidden temporal-relational orientation space:

```
n = (n1, n2, n3), with |n| = 1
```

This is not yet the full 3T sector. It is the smallest reduced object that can be tested against a two-state quantum system.

The vector can be parameterized using two angles:

```
n = (sin(theta) cos(phi), sin(theta) sin(phi), cos(theta))
```

• theta controls how much of the hidden direction points toward or away from the observer's outcome axis.

• phi controls the phase-like orientation around that axis.

• The normalization |n| = 1 means the hidden direction lies on a unit sphere.

5. Candidate projection operator Pi_2A

The candidate projection maps the hidden direction into a two-component complex spinor:

```
Pi_2A(n) = |psi(n)> = cos(theta/2)|0> + exp(i phi) sin(theta/2)|1>
```

Term reminders:

• cos means cosine, a trigonometric function.

• sin means sine, a trigonometric function.

• exp(i phi) means a complex phase. It has magnitude 1 and rotates the amplitude in the complex plane.

• i is the imaginary unit, with i^2 = -1.

• theta/2 appears because two-state quantum spinor geometry uses half-angles.

Plain-language meaning: the hidden 3T orientation is not read directly as a classical direction. It is compressed into a two-state amplitude pair, where theta controls the probability balance and phi preserves interference-capable phase information.

6. Born-rule compatibility check

The Born rule says:

```
P(0) = |alpha|^2
P(1) = |beta|^2
```

From the candidate projection:

```
alpha = cos(theta/2)
beta  = exp(i phi) sin(theta/2)
```

Therefore:

```
P(0) = cos^2(theta/2)
P(1) = sin^2(theta/2)
```

Using the identity below, the probabilities automatically normalize:

```
cos^2(theta/2) + sin^2(theta/2) = 1
```

Important: this is Born-rule compatibility, not a Born-rule derivation. The model produces amplitudes whose squared magnitudes behave correctly, but P = |psi|^2 is still used as the probability-reading rule in this first version.

7. Numerical and constraint checks

Probability normalization means all outcome probabilities must add to 1:

```
P(0) + P(1) = 1
```

Test	Result
Random hidden directions sampled	10000
Maximum normalization error	4.441e-16
Mean normalization error	4.595e-17
Maximum z-axis Born compatibility error	2.220e-16
Maximum random-axis probability range error	0.000e+00

8. Unit rotation compatibility

A physical two-state quantum system must support rotations of measurement basis. In standard quantum mechanics, pure qubit states correspond to points on the Bloch sphere.

In this toy model, rotating the hidden vector n corresponds to rotating the observable spinor |psi>. This is compatibility with ordinary qubit geometry.

```
Hidden rotation:        n -> R n
Observable rotation:    |psi> -> U |psi>
```

• R is an ordinary three-dimensional rotation acting on the hidden direction vector.

• U is a unitary operator acting on the two-state quantum amplitude.

• Unitary means the operation preserves total probability.

Survival result: the proposed projection does not immediately break ordinary qubit rotation structure.

9. No-signalling check

No-signalling means entanglement must not allow controllable faster-than-light communication.

The minimal check uses the Bell state:

```
|Phi+> = (|00> + |11>) / sqrt(2)
```

If the remote observer chooses different measurement bases, the local observer's marginal probability must remain unchanged. For the Bell state, the local reduced density matrix is:

```
rho_A = [[1/2, 0],
         [0, 1/2]]
```

No-signalling test	Value
rho_A from remote Z-basis reading	[[0.5, 0.0], [0.0, 0.5]]
rho_A from remote X-basis reading	[[0.5, 0.0], [0.0, 0.5]]
Maximum marginal difference	0.000e+00

Survival result: hidden relational proximity may be used as an interpretation of correlation only if it does not create controllable observable signalling. This version preserves the no-signalling constraint at the Bell-state marginal level.

10. What Toy Model 2A achieves

- It defines a concrete candidate projection operator Pi_2A.
- It maps a normalized hidden 3D relational orientation into a two-state complex amplitude.
- It preserves probability normalization.
- It is compatible with the Born-rule form for a two-state system.
- It is compatible with ordinary qubit rotation geometry.
- It does not violate the minimal no-signalling Bell marginal check.

11. What Toy Model 2A does not achieve

- It does not derive the Born rule from first principles.
- It does not prove that 3T physically exists.
- It does not derive Hilbert space from 6D geometry.
- It does not solve measurement collapse.
- It does not explain decoherence yet.
- It does not yet connect to black holes, dark matter, or dark energy.
- It does not produce new experimental predictions.

12. Interpretation inside the 6D Projection Framework

```
hidden 3T orientation
      -> projection operator Pi_2A
      -> observable two-state amplitude
      -> Born-compatible outcome probabilities
```

This supports the framework only in a limited sense: a hidden higher-dimensional relational orientation can be compressed into an observer-accessible two-state quantum-like description without immediately breaking standard quantum constraints.

This is the quantum-side analogue of what Toy Model 1D did on the graph-geometry side: it translates a conceptual architecture into a minimal mathematical object and tests whether that object survives basic constraints.

13. First failure risks

- The model may be only a re-description of the standard Bloch sphere, not a new physical derivation.
- The projection currently assumes complex amplitudes rather than deriving why amplitudes must be complex.
- The probability rule is still read using the Born rule rather than derived independently.
- The hidden 3T interpretation must avoid becoming unfalsifiable.
- Extension to multi-particle entanglement will be the real stress test.

14. Next required tests

1. Toy Model 2B: two-particle relational projection and Bell correlations.

2. Toy Model 2C: decoherence as projection-stabilization into one observable 1T record.

3. Toy Model 2D: test whether hidden relational phase can explain interference without violating no-signalling.

4. Toy Model 2E: compare the projection operator with standard Hilbert-space formalism and identify what is merely renamed versus what is structurally new.

5. Only after these tests: return to black holes, horizon entanglement, and dark-sector accessibility.

15. Proposed official close-out statement for 2A

Toy Model 2A successfully defines a minimal reciprocal projection operator from a normalized hidden 3D relational orientation into a two-state quantum amplitude. The model passes first constraints of probability normalization, Born-rule compatibility, qubit rotation compatibility, and a minimal Bell-state no-signalling check. It does not derive Quantum Mechanics or the Born rule. It is a valid starting point for the next quantum-side toy model.

16. Short rebind prompt for continuation

```
We are resuming after V4.2 Toy Model 2A.

2A defined the candidate projection:
Pi_2A(n) = cos(theta/2)|0> + exp(i phi) sin(theta/2)|1>

where:
n = hidden normalized 3T relational orientation
theta = angle to observer's outcome axis
phi = phase-like orientation around that axis

2A passed:
- probability normalization
- Born-rule compatibility
- qubit rotation compatibility
- minimal Bell-state no-signalling marginal check

2A did not derive:
- Born rule
- Hilbert space
- measurement collapse
- decoherence
- physical 3T existence

Next recommended chapter:
Toy Model 2B - Two-Particle Relational Projection and Bell Correlations.

Guardrail:
No claims of deriving Quantum Mechanics. Treat 2A as a compatibility model and a formal starting point.
```